

**Q1.**

The binary operation  $*$  is defined as

$$a * b = a + b + 4 \pmod{6}$$

where  $a, b \in \mathbb{Z}$

- (a) Show that the set  $\{0, 1, 2, 3, 4, 5\}$  forms a group  $G$  under  $*$ .

(5)

- (b) Find the proper subgroups of the group  $G$  in part (a).

(2)

- (c) Determine whether or not the group  $G$  in part (a) is isomorphic to the group  $K = (\langle 3 \rangle, \times_{14})$

(3)

(Total 10 marks)



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